Predictive Modeling for Energy Level and Stability Determination of Organic Photovoltaics

Report Template



CREATING MEANINGFUL EXPERIENCES

Academy of Data Science & AI

Block 1B Machine Learning 2025-26

Breda University OF APPLIED SCIENCES

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1 Business Understanding

1.1 Context and Motivation

Organic photovoltaics (OPVs) offer significant advantages over traditional silicon solar cells: they are flexible, lightweight, and can be manufactured using low-cost printing techniques. However, OPVs currently achieve only 10–15% efficiency compared to silicon's 20–23%.

1.2 Stakeholder Challenge

The primary stakeholder, OMEGALAB, faces a core challenge: traditional material discovery is slow and expensive, with each candidate molecule requiring weeks of synthesis and testing at costs of thousands of euros.

1.3 Project Objectives

This project aims to accelerate material discovery by predicting three critical molecular properties from chemical structure alone:

HOMO energy: Electron-donating ability, affects voltage
LUMO energy: Electron-accepting ability, affects light absorption
Stability classification: Degradation resistance, affects lifespan
The band gap (LUMO minus HOMO) is particularly crucial—optimal efficiency requires values between 1.1 and 1.5 eV.

1.4 Anticipated Challenges

The anticipated challenges include the curse of dimensionality (217 descriptors versus limited training samples) and potential distribution mismatch between training and target molecules.

2 Data Understanding

2.1 Data Sources

Dataset	Molecules	Source
HOPV (Primary)	313	Harvard Organic Photovoltaic Dataset
SQL Database	131	Supplementary database
Kaggle Test Set	200	Competition submission target

2.2 Feature Engineering

SMILES molecular structures were converted to 217 numerical RDKit descriptors capturing molecular weight, electronic properties, topological indices, and functional groups.

2.3 Data Quality Analysis

Data quality analysis revealed several issues:

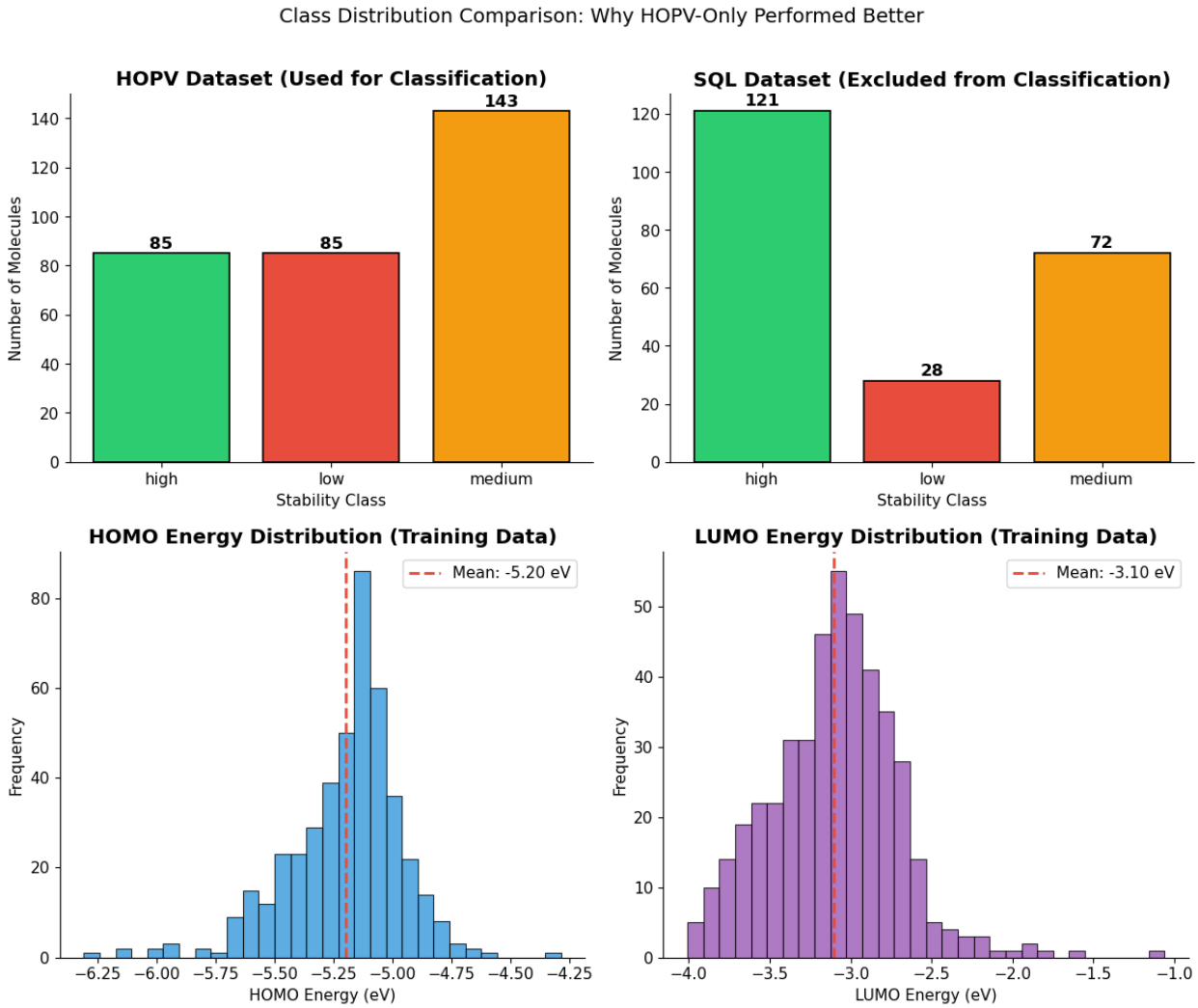
41 constant features providing no predictive information
Missing values in approximately 35% of molecules
3 duplicate SMILES strings

2.4 Distribution Analysis

The most critical insight emerged from distribution analysis:

Dataset	High Stability	Low Stability	Medium Stability
HOPV	85	85	143
SQL	110	27	67

Clustering analysis revealed SQL "low" stability molecules were chemically different from Kaggle molecules, meaning combined datasets degraded classification performance.



3 Data Preparation

3.1 Processing Pipeline

The data preparation pipeline consisted of systematic steps applied consistently to training and test data:

Step	Description	Result
Descriptor Computation	217 RDKit descriptors from SMILES	217 features
Variance Thresholding	Removed constant features	217 → 176 features
Median Imputation	Filled missing values robustly	Complete data
Target Filtering	Dropped rows with missing targets	Clean targets
Standard Scaling	Zero mean, unit variance	Standardized features

3.2 Dataset Selection Decision

For classification, only the HOPV dataset was used—adding SQL data degraded F1 score from 0.90 to 0.71 due to distribution mismatch in the "low" stability class.

4 Classification

4.1 Model Selection

All standard classifiers were evaluated: SVC, Random Forest, Gradient Boosting, KNN, and ensembles. Logistic Regression outperformed complex models due to limited training data—simpler models generalize better under these conditions.

4.2 Best Model Configuration

The best model achieved 0.85 F1 score on both cross-validation and Kaggle leaderboard using:

Algorithm: Logistic Regression
Feature Selection: SelectKBest with f_classif scoring
Features Selected: k = 40
Training Data: HOPV dataset only (313 samples)

4.3 Performance Results

Metric	CV Score	Kaggle Score
F1 (Macro)	0.90	0.85

Predicted Kaggle Distribution:

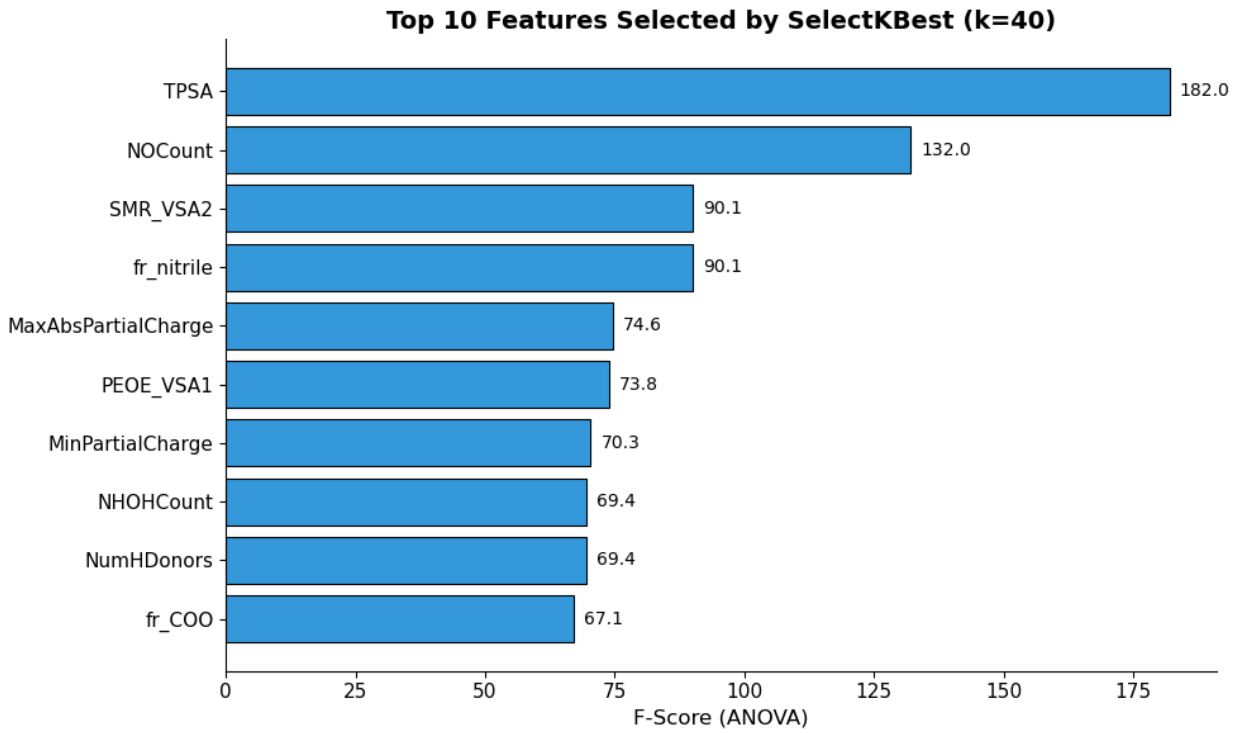
High: 101 (50.5%)
Low: 24 (12.0%)
Medium: 75 (37.5%)
Critically, only the HOPV-trained model could predict the "low" class—models using combined or SQL data predicted zero "low" instances.

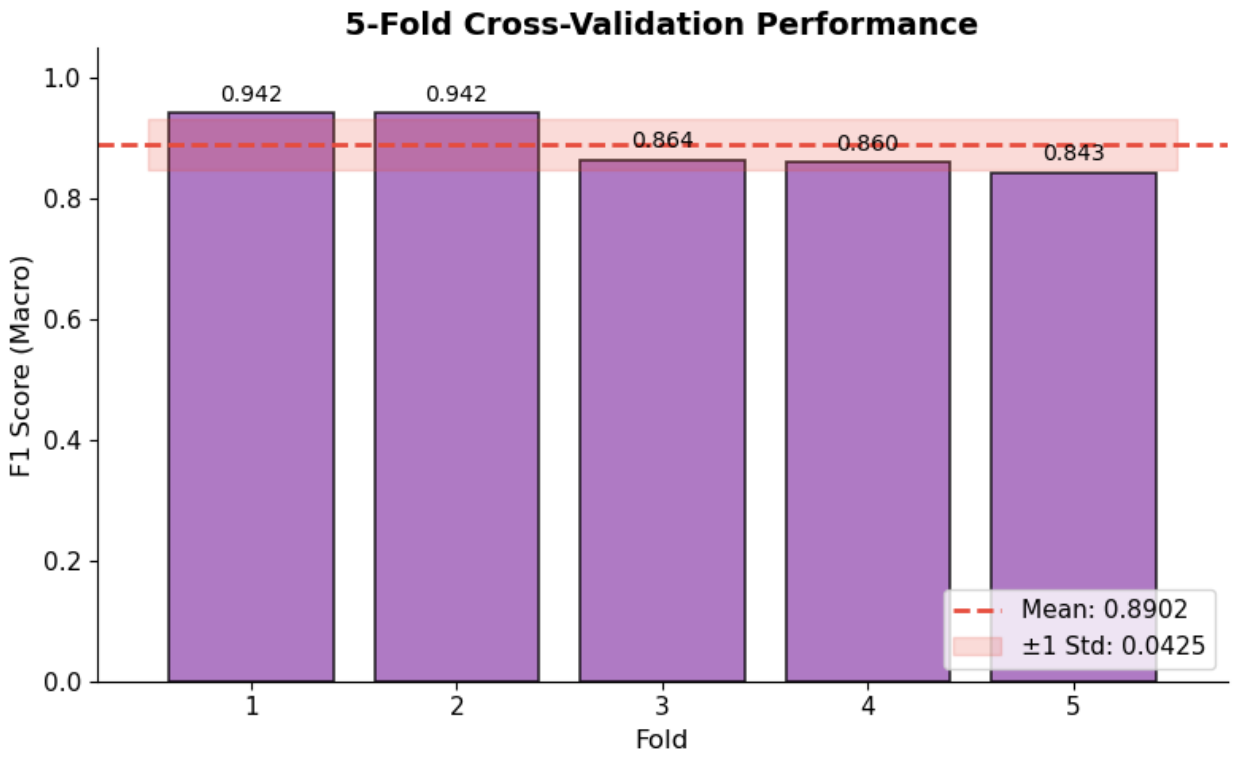
4.4 Top Predictive Features

Rank	Feature	F-Score	Chemical Relevance
1	TPSA	182.0	Polarity
2	NOCount	132.0	Heteroatom content
3	SMR_VSA2	90.1	Surface area
4	fr_nitrile	90.1	Functional groups
5	MaxAbsPartialCharge	74.6	Electronic properties
6	PEOE_VSA1	73.8	Charge distribution
7	MinPartialCharge	70.3	Electronic properties
8	NHOHCount	69.4	Hydrogen bonding
9	NumHDonors	69.4	Hydrogen bonding
10	fr_COO	67.2	Functional groups

These features relate to polarity, heteroatom content, partial charges, and hydrogen bonding capacity—all chemically relevant to molecular stability.

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Top 10 Features Selected by SelectKBest (k=40)
TPSA
NOCount
SMR_VSA2
fr_nitrile
MaxAbsPartialCharge
PEOE_VSA1
MinPartialCharge
NHOHCount
NumHDonors
fr_COO
182.0
132.0
90.1
90.1
74.6
73.8
70.3
69.4
69.4
67.1
0
25
50
75
100
125
150
175
F-Score (ANOVA)


5-Fold Cross-Validation Performance
F1 Score (Macro)
1.0
0.8
0.6
0.4
0.2
0.0
0.942
0.942
0.864
0.860
0.843
Mean: 0.8902
±1 Std: 0.0425
1
2
3
4
5
Fold

5 Regression

5.1 The Sample Selection Problem

Training data contained molecules chemically dissimilar to the Kaggle test set, introducing noise that degraded prediction accuracy. The solution was to selectively identify training molecules most representative of the test distribution using outlier detection methods.

5.2 Sample Selection Methods

Gaussian Mixture Model (GMM): A probabilistic clustering algorithm that models data as a mixture of Gaussian distributions. Training samples from clusters containing Kaggle molecules were selected as chemically relevant.

Local Outlier Factor (LOF): A density-based algorithm measuring how isolated each point is relative to its neighbors. Training molecules identified as "inliers" relative to the Kaggle distribution were selected.

Isolation Forest: Detects anomalies by measuring how easily a point can be isolated through random partitioning. This method was tested but underperformed compared to GMM and LOF.

Interestingly, HOMO and LUMO required different selection methods. LOF worked best for HOMO while GMM was optimal for LUMO, suggesting different distribution characteristics for each target.

5.3 Grid Search Methodology

A comprehensive grid search tested GMM (n_components: 2–5), Isolation Forest (contamination: 0.05–0.2), and LOF (n_neighbors: 10–50). For each method, sample counts, feature counts via SelectKBest, and Ridge regularization strength were systematically optimized.

5.4 Best Model Configurations

Target	CV MAE	Sample Selection	Samples	Features	Ridge α
HOMO	0.0415	LOF (k=30)	100	60	0.5
LUMO	0.0754	GMM (n=4)	100	50	1.5

Combined Performance: CV MAE = 0.0584, Kaggle Score (Average MAE) = 0.09

5.5 Feature Analysis

Top Features for HOMO Prediction:

Feature	Coefficient
SPS	0.161
NumHDonors	0.061
fr_Ar_NH	0.061
NHOHCount	0.061
fr_NH1	0.061
fr_Nhpyrrole	0.061
MolLogP	0.054
HeavyAtomMolWt	0.052

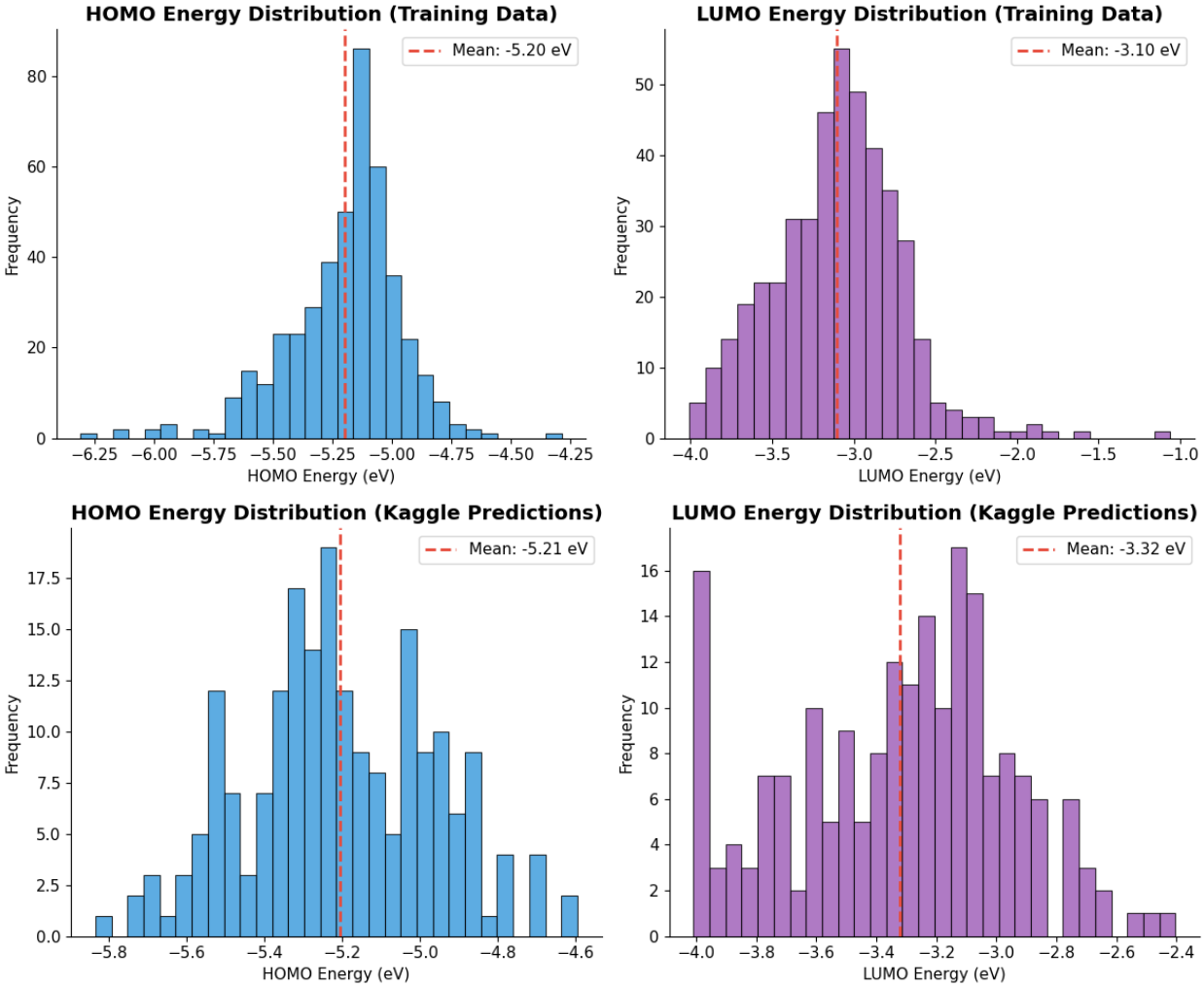
Top Features for LUMO Prediction:

Feature	Coefficient
NumHAcceptors	0.077
fr_pyridine	0.059
NumHeteroatoms	0.056
MinEStateIndex	0.053
fr_NH0	0.045

Shared Features: MolLogP, fr_NH0, and NumHeteroatoms appear in both top 10 lists, indicating these molecular properties influence both orbital energy levels.

5.6 Prediction Summary

Target	Mean	Standard Deviation
HOMO	-5.2064 eV	0.2420
LUMO	-3.3217 eV	0.3661



6 Clustering

6.1 Methodology

K-Means clustering with k=10 was applied to the combined dataset of 779 molecules (579 training + 200 Kaggle) to identify which training molecules are similar to the test set.

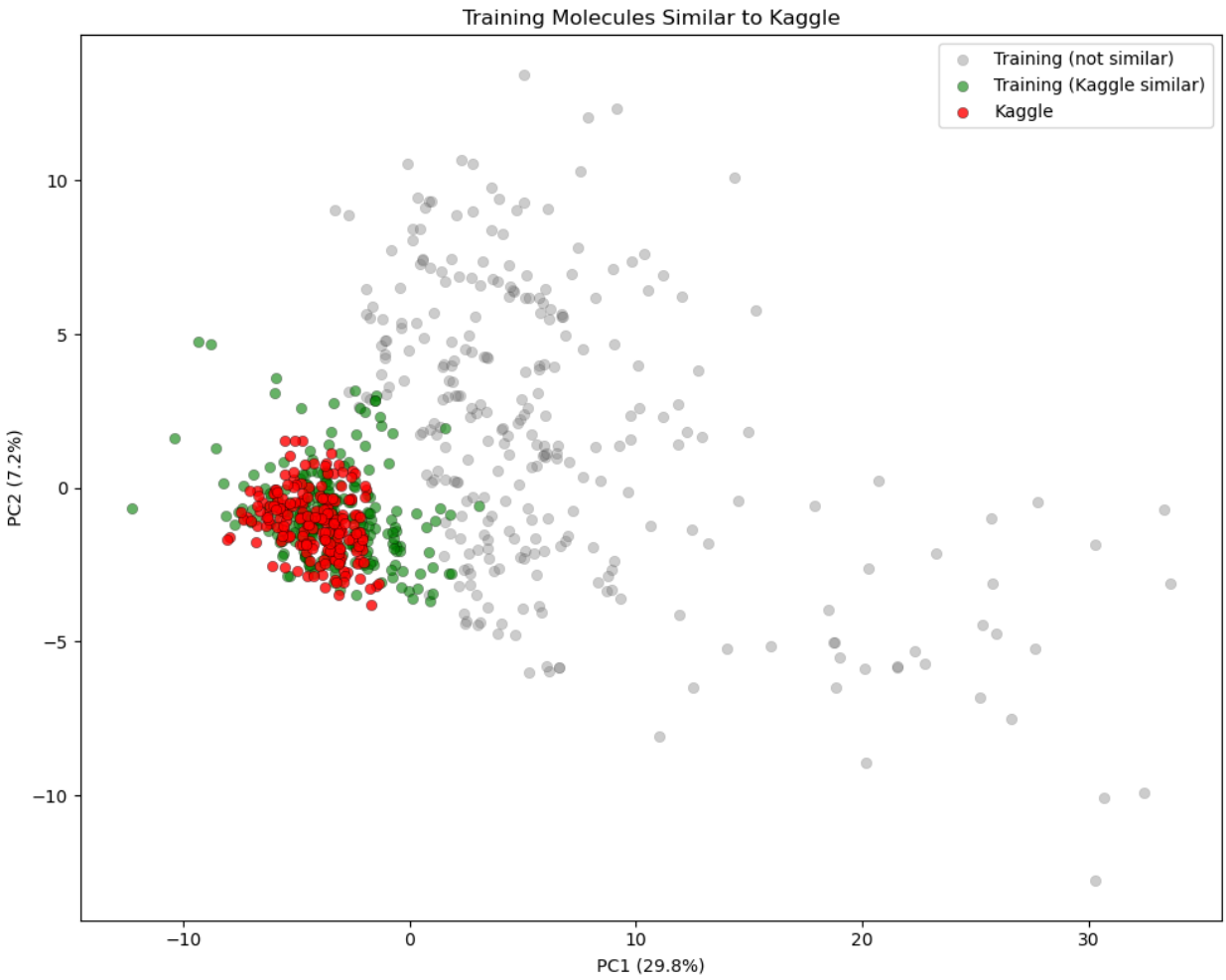
6.2 Cluster Distribution

Category	Clusters	Description
With Kaggle molecules	0 (106), 2 (236), 7 (148)	Chemically similar to test set
Without Kaggle molecules	1, 3, 4, 5, 6, 8, 9	Chemically dissimilar

6.3 Key Findings

Only three clusters (0, 2, 7) contained Kaggle molecules. Training molecules in these clusters were flagged as "Kaggle-similar":

290 out of 579 training molecules (50.1%) share chemical space with the test set
Nearly half of training molecules are chemically dissimilar to Kaggle and may introduce noise
This analysis explains why sample selection methods (LOF, GMM) improved regression performance
The clustering output (kaggle_similarity.csv) enables weighted training and informed validation strategies.



7 Limitations

7.1 Data Constraints

Data Quantity: Only 313 primary training and 131 additional molecules limits learnable pattern complexity.

Domain Shift: Training molecules may not represent the full chemical space of interest.

7.2 Methodological Constraints

Feature Limitations: RDKit descriptors may miss quantum mechanical effects; graph neural networks or molecular transformers could capture more nuanced molecular representations.

CV-Kaggle Gap: Cross-validation estimates tend to be optimistic compared to true generalization performance.

8 Conclusion

8.1 Performance Summary

This project developed machine learning models achieving:

Classification: 0.85 F1 score for stability prediction
Regression: 0.09 MAE for HOMO/LUMO energy prediction

8.2 Key Findings

1. Dataset selection critically impacts performance: HOPV-only outperformed combined datasets for classification due to distribution mismatch

2. Target-specific optimization is essential: Different sample selection methods (LOF vs GMM) are optimal for different targets

3. Chemical similarity matters: Approximately 50% of training molecules are chemically similar to Kaggle test molecules

8.3 Business Impact

For OMEGALAB, these models enable rapid screening of candidate molecules, prioritizing synthesis efforts on promising candidates with predicted optimal band gaps and high stability.

8.4 Future Improvements

Graph neural networks for richer molecular representations
Transfer learning from pretrained molecular transformers
Active learning to efficiently expand training data

References

1. Harvard Organic Photovoltaic Dataset (HOPV). Available at: https://figshare.com/articles/dataset/HOPV15_Dataset/1610063

2. RDKit: Open-Source Cheminformatics Software. Available at: https://www.rdkit.org/

3. Scikit-learn: Machine Learning in Python. Pedregosa et al., JMLR 12, pp. 2825–2830, 2011.



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